

Integrating AI Agents in Enterprise Systems

First Content Module Title in Title Case



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Benefits

"Agentic AI is a new labor model, new productivity model, and a new economic model."

- Marc Benioff, CEO Salesforce

"AI agents represent a multi-trillion dollar opportunity globally."

- Jensen Huang, CEO Nvidia

"AI agents streamline processes, reduce errors, and drive innovation."

- Sundar Pichai, CEO Alphabet

"AI agents enable organizations to be more agile and responsive."

- Satya Nadella, CEO Microsoft



Positive Impact

Customer experience

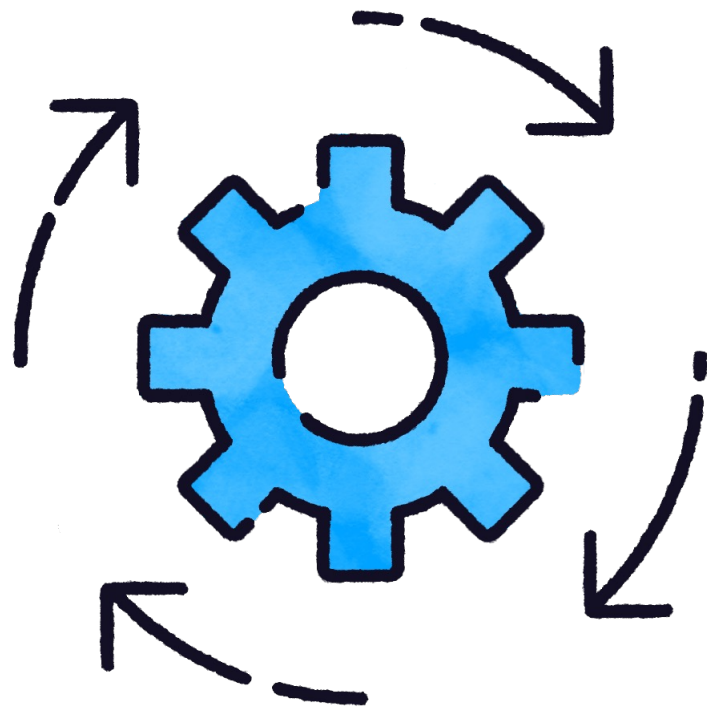
Personalized, responsive 24/7
support

Employee experience

Reduce tedious tasks



Traditional Automation



Robotic process automation (RPA)

Rules-based and ongoing high maintenance



Copilots and virtual assistants

Hallucinations, reactive, constant prompting required

AI Agents

Autonomy

Learn and adapt with minimal prompting

Less maintenance than RPA

More proactive than Copilots



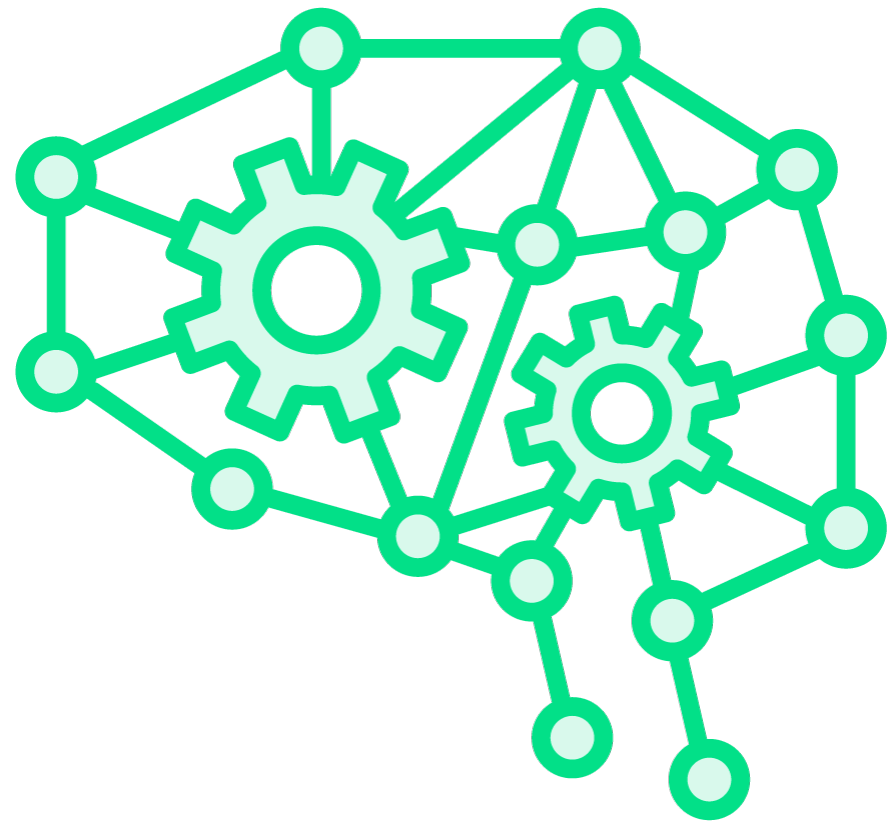
Multi-agent Collaboration

Multi-step reasoning

**Different agents
handle specialized
tasks**

**Deeper insights and
scalable automation**





Traditional apps: heavy user interaction (data entry, clicks)

AI agents: simple UX (natural language input)

Work happens behind the scenes

Sales Rep Example:

- Asks Slack agent: "Prepare briefing for Client John"
- Agent gathers CRM data, emails, support tickets, and analytics
- Provides summary—no app switching needed



Business Model Transformation

Per-agent pricing

**Outcome-based
pricing (e.g., ticket
resolutions)**

**Usage-based pricing
(API calls, tokens,
time)**



Types of AI Agents for Business Workflows





Cloud

**Popular approach for business organizations.
They may use multiple cloud platforms.**



Evaluation of Cloud Platforms

Benefits

- Scalability and elasticity**
- Managed services and reduced operational overhead**
- Global reach and low latency**
- Integration with advanced AI and ML tools**
- Enhanced security and compliance**

Drawbacks

- Cost management challenges**
- Complexity of services and learning curve**
- Vendor lock-in risks**
- Data privacy and compliance concerns**
- Dependence on internet connectivity**



Vendor-specific Platforms

Salesforce's Agentforce

ServiceNow's Now Assist



Evaluation of Vendor-specific Platforms

Benefits

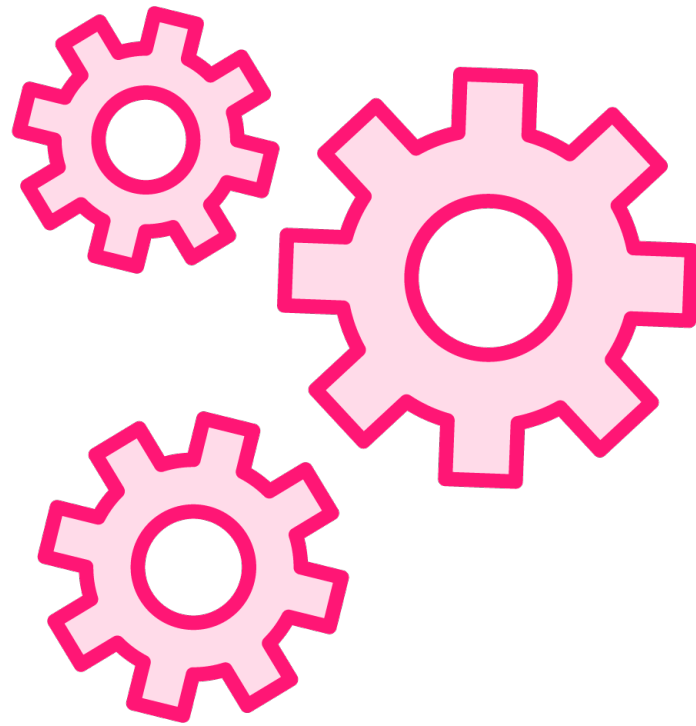
- Ease of adoption**
- Data**
- Seamless integration**
- Specialized solutions**
- Rapid deployment**
- Ongoing support and updates**

Drawbacks

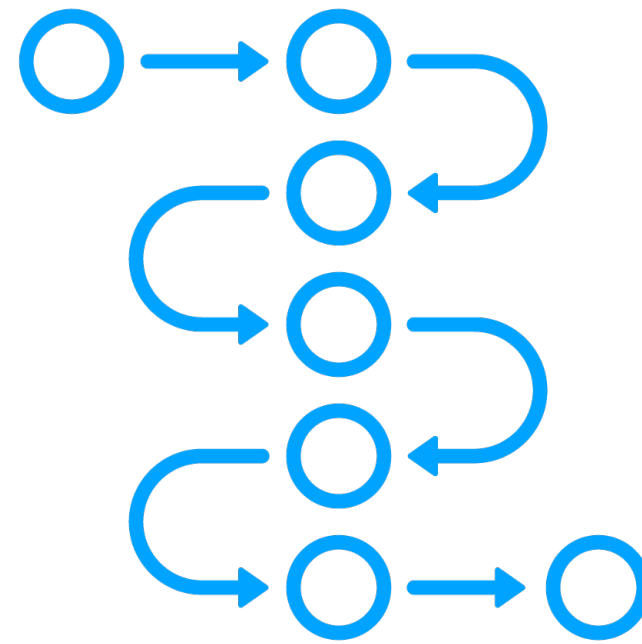
- Limited flexibility**
- Third-party integration**
- Cost considerations**
- Vendor lock-in**
- Data privacy and compliance concerns**



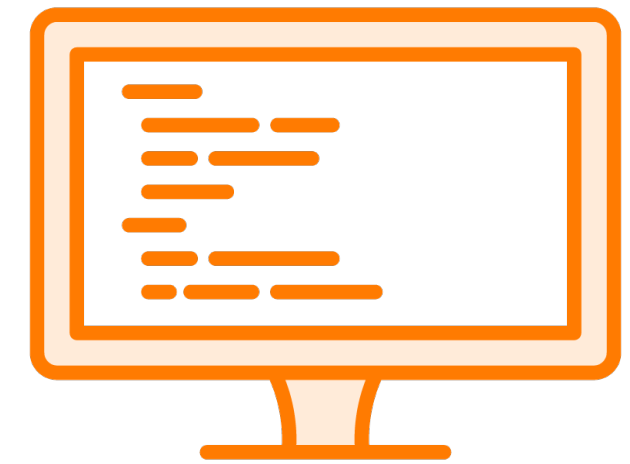
Open Source Frameworks and Libraries



AutoGen



LangGraph



CrewAI

Evaluation of Open Source Frameworks and Libraries

Benefits

Customization and flexibility

Community support and collaboration

Cost-effectiveness

Transparency and control

Drawbacks

Technical expertise required

Maintenance and support responsibility

Licensing issues

Quality

Breakage



Building Native AI Agents



Customization



More time and costs





AI Agents Across Departments





Scale

Without out, there can be data siloes and fragmentation.





HR



**Automated recruitment
Employee onboarding
Employee support**



Finance



**Fraud detection
Automated financial reporting
Expense management**



Marketing



**Personalized campaigns
Content creation
Market analysis**



IT support



**Automate the helpdesk
System monitoring
Automated software deployment**



Planning and Adoption



Define strategic objectives and use cases

Identify business challenges and opportunities. Focus on areas where AI agents have the biggest impact.



Buy-in

Ideally, you want buy-in from senior executives.



Assess Data Infrastructure and Readiness

Data framework

**Data governance
policies**

Autonomy



Other Key Steps for Implementation

**Choose the right
technologies and
platforms**

**Develop ethical
guidelines**

Pilot and iterate

**Scale and integrate
across departments**

**Monitor, evaluate,
and optimize**



Demo: Using CrewAI to Create an AI Marketing Agent



Marketing Agents

Trend analyst

Content creator

**Enagement
optimizer**



GitHub repo: <https://github.com/ttaulli/ai-agents-course>
Folder: social_media_marketer



CrewAI installation: <https://docs.crewai.com/installation>



OpenAI API: <https://platform.openai.com/docs/overview>

